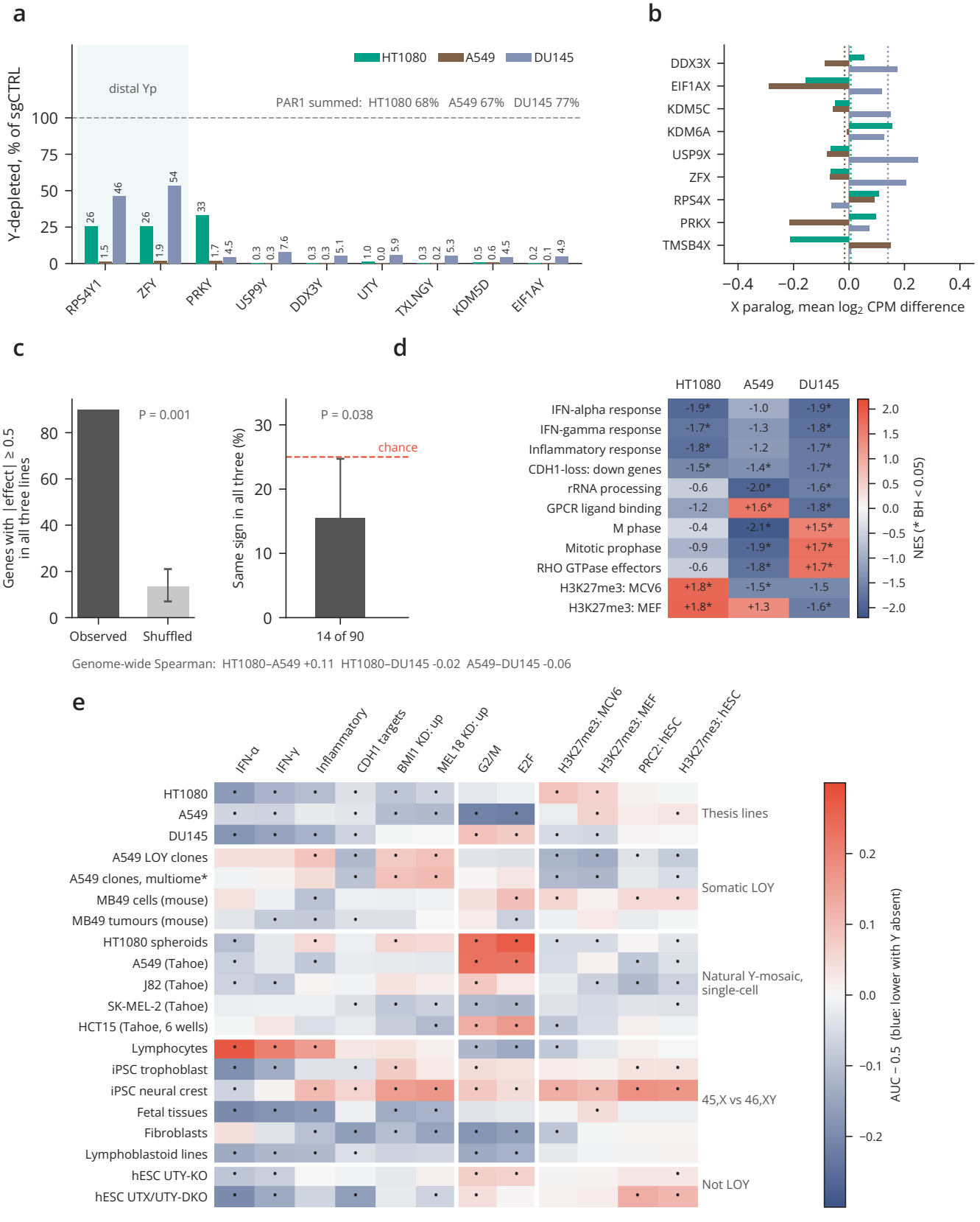

a
HT1080
A549
DU145
distal Yp
PAR1 summed: HT1080 68% A549 67% DU145 77%
Y-depleted, % of sgCTRL
RPS4Y1
ZFY
PRKY
USP9Y
DDX3Y
UTY
TXLNGY
KDM5D
EIF1AY
b
DDX3X
EIF1AX
KDM5C
KDM6A
USP9X
ZFX
RPS4X
PRKX
TMSB4X
X paralog, mean log2 CPM difference
c
P = 0.001
Genes with |effect| ≥ 0.5 in all three lines
Observed
Shuffled
P = 0.038
chance
Same sign in all three (%)
14 of 90
d
HT1080
A549
DU145
IFN-alpha response
IFN-gamma response
Inflammatory response
CDH1-loss: down genes
rRNA processing
GPCR ligand binding
M phase
Mitotic prophase
RHO GTPase effectors
H3K27me3: MCV6
H3K27me3: MEF
NES (* BH < 0.05)
Genome-wide Spearman: HT1080–A549 +0.11 HT1080–DU145 -0.02 A549–DU145 -0.06
e
IFN-α
IFN-γ
Inflammatory
CDH1 targets
BMI1 KD: up
MEL18 KD: up
G2/M
E2F
H3K27me3: MCV6
H3K27me3: MEF
PRC2: hESC
H3K27me3: hESC
HT1080
A549
DU145
A549 LOY clones
A549 clones, multiome*
MB49 cells (mouse)
MB49 tumours (mouse)
HT1080 spheroids
A549 (Tahoe)
J82 (Tahoe)
SK-MEL-2 (Tahoe)
HCT15 (Tahoe, 6 wells)
Lymphocytes
iPSC trophoblast
iPSC neural crest
Fetal tissues
Fibroblasts
Lymphoblastoid lines
hESC UTY-KO
hESC UTX/UTY-DKO
Thesis lines
Somatic LOY
Natural Y-mosaic, single-cell
45,X vs 46,XY
Not LOY
AUC − 0.5 (blue: lower with Y absent)